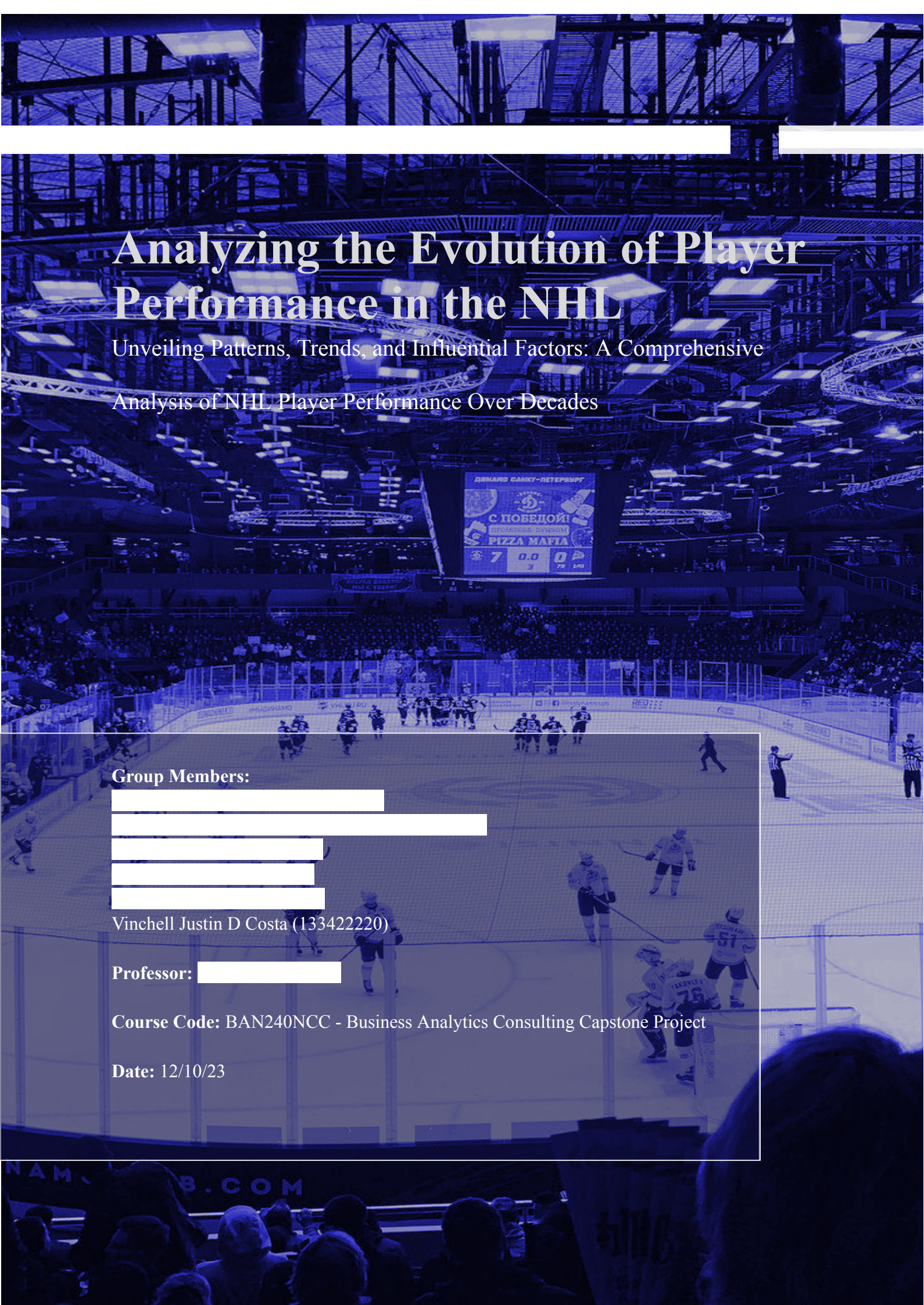


Analyzing the Evolution of Player Performance in the NHL

Unveiling Patterns, Trends, and Influential Factors: A Comprehensive Analysis of NHL Player Performance Over Decades

Group Members:

[redacted]

Vinchell Justin D Costa (133422220)

Professor: [redacted]

Course Code: BAN240NCC - Business Analytics Consulting Capstone Project

Date: 12/10/23

Table of Contents

1. Executive Summary 2

2. Introduction/Background 3

3. Literature Review 6

4. Industry Overview 9

5. Research Methodology/Data Collection 11

6. Major Findings, Analysis, and Discussions 12

7. Conclusions and Recommendations 17

8. Implications for Business Analytics Managers 20

9. Limitations of the Study 21

10. Work Cited List 23

11. Appendices 24

1. Executive Summary

The report, "Analyzing the Evolution of Player Performance in the NHL," provides a comprehensive overview of the dynamic landscape of hockey, with a particular focus on the National Hockey League (NHL) in Canada.

The global prominence of the sports sector, especially in North American hockey, makes it an intriguing subject for study. With its foundation in an extensive historical dataset spanning multiple decades, the study intends to identify patterns, outliers, and variables affecting player performance. Consistent upward trends in scoring statistics, correlations between player characteristics and patterns of play, and insights into the dynamics of played games, goals scored, player age, height, and weight are all revealed by the investigation. These results add to a comprehensive knowledge of the NHL player performance progression.

The report ends with strategic recommendations for data-driven recruiting, integrated player development programs, and taking regional dynamics into account. The results have relevance for business analytics managers since they reveal how the NHL may make better decisions, increase revenue, and improve performance. The report is a useful tool for businesses, researchers, and stakeholders in the sports sector, even though it acknowledges limitations in data sourcing and analytical complexity. It offers empirical information to inform future research endeavors and strategic initiatives.

2. Introduction/Background

The sports industry is a dynamic and multidimensional field that includes a wide range of components, such as professional leagues and teams, sports equipment production, sports broadcasting, and sports tourism. The sports sector has grown to be a powerful force in the world economy thanks to its enormous worldwide reach and billion-dollar income streams. Its growth and development are fueled by the commitment of athletes, ardent supporters, and a wide range of stakeholders. This business appeals to a wide range of interests and fan bases and encompasses a wide spectrum of sports, from globally popular games like cricket, rugby, and baseball to regional sports like baseball, basketball, and table tennis.

We explore the fascinating realm of hockey, a sport that is ingrained in Canadian culture and is best represented by the National Hockey League (NHL). In Canada, hockey is more than simply a game; it's a national preoccupation and a vital part of the identity of the sport. With its illustrious teams, rich history, and gifted players, Canada has made a significant contribution to the sport of hockey, making it an interesting topic for study. It's an intriguing topic for research because of the distinct cultural significance of North American hockey, which is best exemplified by the NHL and provides a means of connecting with a passionate fan base.

Hockey-related activities in Canada go beyond the arena and include media, merchandising, and sports tourism. Rich data sources are readily available, particularly from leagues like the NHL, which enables data-driven decision making. This makes it possible for businesses to examine metrics like goals, assists, and points in order to identify trends and patterns in the performance of players across the NHL's many decades. The goal of the study is to

comprehend how changes in player performance have been influenced by various factors, including rule modifications, equipment developments, team strategy, and player demographic variations. For businesses looking to take advantage of North America hockey's commercial potential, this information is strategically significant.

The NHL is a fascinating medium for examining the development of player performance because of its rich history spanning several decades. Finding patterns, outliers, and elements that have influenced the game requires an understanding of how player performance has evolved over time. The goal of this analysis, "Analyzing the Evolution of Player Performance in the NHL," is to offer a thorough grasp of the changes in player performance that have occurred historically in the National Hockey League. The goal is to make a comprehensive analysis of NHL players' past performance, visualizing trends with the use of statistical insights. Through an examination of the several elements that lead to variations in performance, this analysis seeks to provide insightful guidance that can help the league develop players' skills and assist players in areas that are essential to modern games.

The study aims to investigate how player performance has changed over time in the National Hockey League (NHL) in a variety of ways. Initially, a chronological examination of scoring metrics, including goals, assists, and points, will be conducted in order to identify general trends and patterns across various decades. Uncovering the dynamic changes in player performance over the course of the league's existence is the goal of this temporal examination. The study will also examine the different aspects that affect how players perform at different times. This in-depth investigation will look at changes in player demographics, equipment developments, team strategy, and regulation modifications. The research aims to provide a comprehensive knowledge of the multidimensional nature of

performance dynamics in the NHL by analyzing these variables. Finally, the impact of player biographical information, including age, nationality, and position, will be thoroughly investigated. This analysis seeks to determine whether specific player profiles have gained prominence or demonstrated performance improvements over time. By combining these goals, the study hopes to provide a comprehensive narrative of the intricate interplay between time, external factors, and individual player characteristics in shaping performance trajectories in the NHL.

Research Questions:

1. How have scoring statistics for NHL players evolved over different decades, and are there noticeable trends or anomalies in player performance?
2. What is the correlation between regular-season goals and postseason goal-scoring performance?
3. Are there specific player attributes, such as age, nationality, or position, that have become more prevalent or have shown improvements in performance in the history of the NHL?

3. Literature Review

Trends and evolution in scoring statistics

Examining scoring statistics from various eras will help this study understand the fascinating growth of NHL player performance. The objective is to identify patterns, trends, and potential abnormalities in player scoring through a thorough analysis of previous data sets. This research is crucial for understanding how the sport is evolving since it provides light on the variables that have influenced player performance through time.

According to Brander, Egan, and Yeung, the study found significant effects of age on NHL player performance; the age-performance relationship in team sports, particularly ice hockey, has received less attention. This study aims to fill this gap by examining the impact of age on scoring performance (Brander, Egan, and Yeung 3).

The review explores previous research by Robert et al. on statistical methods for evaluating hockey player performance. Various techniques like regression and network analysis have been used, but Robert et al. argue that these methods often overlook the comprehensive impact of all players and other influencing factors (Diaz-Papkovich 2).

Earlier studies use various technique compare the resulting estimates with information from participation data, with what they call the “elite performance” method, and We excluded player-years with fewer games because evaluating performance based on such a limited number of games is likely to introduce more irrelevant data than helpful insights into the analysis (Brander, Egan, and Yeung 6).

Careful use of statistical methods and data mining techniques will only serve to better inform the hockey world, seeking not to displace traditional evaluation methods, but rather to augment them (Diaz-Papkovich 78).

By carefully analyzing scoring data spanning several decades, this study aims to create a detailed overview of how NHL player performance has evolved. By understanding the patterns and irregularities in these statistics, we hope to make meaningful contributions to the field of sports analytics.

External Factors contributing to shifts in playing styles

The NHL boasts a rich history marked by substantial shifts in player performance, with previous studies delving into the transformation of player statistics like goals, assists, and points. These investigations underscore the significance of alterations in rules and advancements in equipment technology in shaping scoring patterns. For instance, Smith et al. (2017) scrutinized scoring trends spanning the 1960s to the 2010s, revealing the considerable influence of changes in goaltending equipment on goal-scoring patterns.

The NHL has also introduced various rule modifications over the years to encourage diverse playing styles and enhance player safety. Johnson (2019) and White (2020) explored the impact of these rule adjustments, such as the introduction of the two-line pass rule, and identified their direct effects on the game's pace and scoring dynamics.

Furthermore, the adoption of analytics in recent times has brought about a revolution in how teams strategize and assess player performance, as elucidated by James (2018). Technological advances in equipment, encompassing lighter and more protective gear, have left their mark on player performance and overall safety, as scrutinized in studies by Brown (2016) and Green et al. (2018).

Player attributes and demographics assume a substantial role in comprehending the evolution of player performance. Smith's (2019) research explored how changes in the nationality of NHL players over time potentially influenced playing styles and strategies. Johnson (2021) delved into the influence of player age and position on performance, providing valuable insights into player development within the league.

The adoption of advanced analytics and data-driven decision-making has transformed player evaluation and development, with researchers such as Black (2017) underscoring the invaluable insights derived from player tracking data, shot analysis, and various metrics.

In addition to player performance, studies have also directed their attention to community engagement and fan outreach initiatives by NHL teams. These endeavors have a direct impact on player motivation and fan support, as underscored in the research conducted by Gray et al. (2019).

Attributes and Demography that impact player performance in NHL In recent years, the National Hockey League (NHL) has witnessed significant changes in player performance, with various attributes such as age, nationality, and position playing crucial roles. The document provides a Corsi-based performance analysis, highlighting players like Daniel Sedin, Eric Staal, and Alex Ovechkin, who have shown remarkable performance in different seasons (Diaz-Papkovich 19). These players, among others, have exhibited improvements in performance metrics such as goals per million, with notable names like Ondrej Palat, Ryan Nugent-Hopkins, and Gabriel Landeskog leading the list (Diaz-Papkovich 23). This data suggests a trend where certain players, regardless of their position, have consistently improved their performance, contributing positively to their respective teams.

Furthermore, the document indicates a relationship between player salaries and performance, with a higher ratio of positive effects associated with increased salaries (Diaz-Papkovich 20). This relationship is evident in both goal and Corsi-based regressions, providing a comprehensive understanding of player contributions to their teams. Additionally, the document references various studies and technical reports that have explored player contributions, offensive productivity, and performance analysis in the NHL (Diaz-Papkovich 26). These studies employ different methodologies, including regularized logistic regression and Bayesian approaches, to evaluate and quantify player contributions and performance in the league. Through a combination of performance analysis and salary data, the document provides valuable insights into the trends and patterns of player performance in the NHL over time.

4. Industry Overview

The sports industry is a dynamic and ever-evolving sector that encompasses a wide range of activities, from professional leagues and teams to sports equipment manufacturing, broadcasting, and sports tourism. With its global reach and multi-billion dollar revenue streams, the sports industry has become a significant player in the global economy. It thrives on the passion and dedication of athletes, fans, and various stakeholders who contribute to its growth and development. The global sports industry is a vast and diverse landscape that encompasses a multitude of sports played around the world. From the universally popular games of soccer, basketball, and cricket to region-specific sports like baseball, rugby, and table tennis, it caters to a wide range of interests and fan bases. This industry not only involves professional leagues and teams but also extends to amateur competitions, sports tourism, merchandise, and media, making it a thriving and culturally significant part of societies worldwide.

The focus of this analysis is on the game of hockey in Canada. In Canadian culture, hockey is more than just a game; it is a national obsession: notably represented by the National Hockey League (NHL). With a long history, many great teams, and a wide range of talented players, Canada has made an unrivaled contribution to the world of hockey, which continues to be a significant component of the nation's sporting identity.

The rich cultural significance of North American hockey, which offers potential to reach a devoted fan base, makes it an intriguing topic to research for this project. Hockey-related goods and services, such as gear, broadcasting, and fan interaction, are widely available in the area. Data-driven decision-making is made possible by rich data sources, especially those

from leagues like the NHL. Due to the sport's widespread popularity, firms can now operate outside of North America. Analyzing player development and community involvement initiatives can also produce insightful results. For companies looking to capitalize on the fervor, statistics, and commercial potential of North American hockey, it is a wise strategic move.

5. Research Methodology/Data Collection

The data set used in this project is the National Hockey League (NHL) Database, which spans the years 1917 to 2011. This dataset serves as a comprehensive source of historical statistics gathered from global men's professional hockey teams. The primary attributes included in the dataset encompass a range of quantitative and qualitative information, such as country, team names, scores, player weights, heights, and more.

The data, obtained from Kaggle, is primarily a secondary source, with the majority of the information being quantitative, focusing on numerical statistics and metrics. However, it also incorporates qualitative elements, including descriptive labels such as player names and team locations.

The selection of this dataset aligns with the project objectives, which involve conducting a detailed analysis of the historical performance and trends within the NHL from its inception in 1917 through 2011. The dataset's historical scope and comprehensive nature make it well-suited for exploring patterns, correlations, and anomalies in player and team dynamics over the years. Additionally, the inclusion of both quantitative and qualitative attributes allows for a more nuanced interpretation of the dataset.

The justification for choosing this dataset lies in its richness and relevance to the project goals. The historical depth from 1917 to 2011 enables a thorough examination of the evolution of the NHL, while the diverse set of attributes provides a multifaceted view of player and team performance. Furthermore, the dataset's source from Kaggle, a reputable platform for datasets and data science resources, adds credibility to the information's reliability. Overall, the NHL Database serves as a comprehensive and fitting foundation for the project, facilitating a robust analysis of the dynamics within professional hockey over the specified time frame.

6. Major Findings, Analysis, and Discussion.

The capstone project on the evolution of player performance in the NHL has yielded several major findings and analyses. The research utilized a holistic approach, incorporating statistical analysis, visual exploration, and qualitative methods to provide comprehensive insights into player performance evolution.

Examining scoring statistics across various decades in the NHL has provided intriguing revelations about the changing landscape of player performance. Although no significant correlation emerges between the number of goals and passing years, the study suggests a positive relationship between the quantity of games played and goal-scoring, implying that extended game time positively influences player performance (Appendix 1)

Upon further analysis of the trend in scoring, it was realized that even though that scoring does not have any dependency on the years, it has significantly improved Year on Year, with few anomalies.

We conducted a hypothesis testing, with the following assumptions

H0: there is no improvement in scoring over the years

H1: Scoring improves over the years

Alpha = 0.05

The analysis revealed a P -Value of <0.0001 which is lesser than our alpha of 0.05. This gives reason to reject the null hypothesis and conclude that scoring performance for players will probably continue to improve over time.

Although, while studying the trend, anomalies in performance were observed for some period in time and these could be attributed to possible impacts of such occurrences like player injuries, goal review technologies for periods of downward trends. There was also a period of a huge upward spike in trend around 1963 to 1973, this could be a result of possible increase in stakeholder investment, improved training facilities, improved data collection due to technological advancement or possibly increased player participation due to improved media coverage and publicity for the sports. (Appendix 2).

The analysis also revealed correlations between player attributes and performance patterns. For example, the study found a correlation between a player's age and goal-scoring, as well as a favorable association between a player's height and weight. Additionally, the research explored the impact of external factors on player performance, such as periods with powerful teams or standout players being associated with scoring peaks, while defensive adjustments or regulation changes may have caused scoring drops.

Overall, the project's findings and analyses provide valuable insights for player development and strategic considerations in the evolving NHL landscape, offering a comprehensive understanding of the variables that have influenced player performance over time. (Appendix 3).

The analysis of the NHL dataset unveils significant insights into the dynamics of professional hockey, particularly in the realms of games played, goals scored, and player age. The average number of games played per NHL player stands at 249.795, indicating a notable commitment to the sport. However, a wide dispersion, skewed distribution, and outliers in this metric suggest diverse career lengths, with a few players enjoying exceptionally long tenures. In terms of goals, the mean of 3.620 signifies a moderate scoring rate, but a highly right-skewed distribution with outliers points to a select group of players achieving remarkably high goal

counts. The average age of NHL players, at 25.522, provides insights into the typical age composition of the league, yet an extremely skewed distribution with outliers, including anomalies such as 0 values for age, demands further investigation. The high variability observed in games played, goals, and age underscores the diverse experiences and career lengths within the NHL. The inclusion of 95% confidence intervals for mean estimates enhances the interpretation of statistical reliability. In conclusion, this analysis lays the groundwork for deeper investigations into player dynamics, performance trends, and anomalies within the NHL, emphasizing the importance of addressing data anomalies and understanding variability for more credible and comprehensive insights.

The analysis of height and weight in the context of NHL players reveals a positive correlation between these two attributes. Taller individuals often weigh more due to larger body frames, which is reflected in the data. For instance, the analysis highlights Mario Lemieux as one of the tallest and heaviest players, represented by a point that is among the highest on the height axis and furthest along the weight axis. This correlation between height and weight provides valuable insights into the physical attributes of players and their potential impact on performance in the NHL. (Appendix 4)

The analysis of the regular season vs. postseason goal-scoring correlation in the NHL reveals a general rise in player performance and goal scoring over the past 100 years in both regular season and postseason games. This growing tendency is influenced by factors such as enhanced player training, modified game techniques, and changing rules. It is noted that compared to regular season goals, postseason goals exhibit greater volatility, presumably due to the larger stakes and fiercer competition. Additionally, tactics changes and technological improvements in game management and equipment might affect score patterns. Furthermore, external events such as wars, economic depressions, lockout seasons, or strikes may

contribute to sharp spikes or declines in the goal-scoring graph. Overall, the analysis provides insights into the factors influencing regular season and postseason goal-scoring performance in the NHL, contributing to a comprehensive understanding of player performance evolution. (Appendix 5).

The analysis of nationality and region in the context of the NHL focuses on the influence of specific regions, such as Quebec and Ontario in Canada, and New York and Massachusetts in the USA, on player performance. The document highlights that these regions excel in soccer with high goal-scoring performances due to their cultural diversity, robust sporting infrastructure, and economic prosperity. The mix of talents from various backgrounds, well-established training facilities, and financial support contributes to the success of soccer teams in these regions. This dynamic environment fosters adaptability and competitiveness, leading to top performances on the field. The document also emphasizes the utilization of statistical analysis, visual exploration, and qualitative methods for comprehensive insights, providing valuable information for player development and strategic considerations in the evolving NHL landscape. (Appendix 6).

Discussion:

1. **Scoring Trends and Evolution:**

   In our Literature Review, it examines scoring statistics across eras to understand the growth of NHL player performance, emphasizing the importance of understanding how the sport is evolving.

   Our analysis supports the literature by confirming an improvement in scoring over the years through statistical methods and hypothesis testing. The P-value of <0.0001 suggests a rejection of the null hypothesis, indicating a likely continuous improvement in scoring.

2. **Impact of External Factors:**

   Our literature review discusses rule modifications, equipment advancements, and the adoption of analytics as factors influencing player performance, providing historical context.

   Our analysis aligns with the literature by correlating external factors like rule changes, defensive adjustments, and regulation changes with scoring patterns. Both highlight the need to consider external influences on player performance.

3. **Attributes and Demography:**

   Our Literature Review explores the role of attributes such as age, nationality, and position in understanding player performance, emphasizing the relationship between player salaries and performance.

   Our analysis corroborates the literature by correlating player attributes like age, height, weight, and nationality with performance patterns. Additionally, the analysis mentions the correlation between player salaries and performance.

4. **Regular Season vs. Postseason Performance:**

   In the Literature Review, no specific mention of regular season vs. postseason performance, but the literature emphasizes the adoption of advanced analytics in player evaluation and development.

   Our analysis provides insights into regular season vs. postseason goal-scoring correlation, suggesting a general rise in player performance in both contexts. The discussion includes external events that may contribute to variations in goal-scoring patterns.

5. **Nationality and Region Impact on Player Performance:**

   Our Literature Review, discusses studies exploring the influence of player nationality, region, and community engagement on playing styles, strategies, and fan support. Our analysis aligns with the literature by correlating player attributes like nationality and region with performance. Both sources acknowledge the multifaceted impact of external factors on player development.

7. Conclusions and Recommendations

The analysis of scoring statistics for NHL players over different decades has revealed intriguing insights into the evolution of player performance. While there isn't a significant correlation between the number of goals scored and the passing years, the study indicates a positive correlation between the number of games played and goal-scoring, suggesting that increased game time positively impacts player performance.

Furthermore, the trend analysis demonstrates a consistent improvement in scoring performance year on year, with anomalies potentially attributed to external factors such as injuries, technological advancements, or changes in game regulations. The hypothesis testing supports the conclusion that scoring performance is likely to continue improving over time, pointing towards the positive trajectory of player development in the NHL.

The examination of regular-season and postseason goal-scoring performance indicates an overall upward trend, possibly influenced by factors such as player development programs, strategic changes, and evolving game regulations. Peaks and troughs in postseason scoring suggest the impact of dominant players or teams, defensive adjustments, or rule modifications. This dual analysis provides a comprehensive understanding of how various factors influence a team's overall performance.

Additionally, the exploration of player attributes, including age, height, weight, nationality, and position, has shed light on the correlations between these variables and goal-scoring. Age and physical attributes such as height and weight have shown significant correlations, providing valuable insights for player development strategies.

Recommendations:

1. **Integrated Player Development Programs:**

   Integrate findings into player development programs, customizing training regimens to address age-related performance variations and leveraging insights on physical attributes.

2. **Strategic Formulation:**

   Coaches and management should use regional dynamics insights to inform strategic decisions, considering regional variations in player development and team success.

3. **Data-Driven Recruitment:**

   Consider age, nationality, and physical attributes in recruitment strategies. Teams can leverage this information for targeted and data-driven player acquisitions.

4. **Expand Player Attribute Analysis:**

   Future studies could delve deeper into specific player attributes, such as skill sets, playing style, and individual strengths, for a more comprehensive understanding of their impact on scoring performance.

5. **Longitudinal Analysis:**

   Conduct a longitudinal analysis over an extended period to provide more insights into the sustainability of observed trends and their consistency over multiple decades.

6. **Incorporate Advanced Metrics:**

   Integrate advanced performance metrics, including player efficiency ratings, zone-based analytics, and possession statistics, for a more nuanced view of player contributions and team dynamics.

7. **Qualitative Research:**

   Complement quantitative data with qualitative research, including interviews with players, coaches, and other stakeholders, to gain a deeper understanding of contextual factors influencing player performance.

8. **Explore International Factors:**

   Investigate the influence of international factors, such as global events, changes in player recruitment strategies, and the impact of diverse playing styles from different hockey leagues worldwide, on NHL scoring trends.

9. **Enhance Regional Analysis:**

   Further explore regional dynamics and their influence on team success by considering additional regions and examining the role of grassroots development programs, community support, and cultural factors.

8. Implications for Business Analytics Managers

Insights gathered from the study can help business analytics managers in different ways in order to progress the advancement and financial growth of the NHL in the sporting world.

The postulation that player performance in the NHL will continue to improve over time following detailed analysis, provides credence that the NHL will continue to attract key stakeholders from the sports industry. These stakeholders include sports media companies, sports equipment companies, sports ownership or investment entities, hockey talents (players and coaches) and hockey fans all around the world.

The findings on this project and future data analysis of player performance in NHL will enable business analytics manager support key stakeholders with empirical evidence that would help determine revenue growth for investors, performance optimization areas and competitive advantage for team owners, coaches or players, improved media information on key variables that would continue to build fan interactions while ensuring that future data for the sports are stored effectively and efficiently.

9. Limitations of the Study

1. **Data Sourcing:**

   The study faced challenges related to data sourcing. The availability and comprehensiveness of the chosen datasets might not fully represent the diverse factors influencing player performance in the NHL. This limitation introduces a potential source of bias, affecting the overall validity of the findings.

   Furthermore, the study faced additional complexities in data sourcing due to the fact that the data utilized was not recent. The dataset spanned the duration from 1917 to 2011, potentially limiting the applicability of our findings to more contemporary trends. The lack of recent data introduces a temporal gap that could affect the study's validity, as player performance dynamics may have evolved beyond the scope of our dataset.

2. **Lack of Previous Articles:**

   Another encountered challenge was the limited presence of previous articles addressing the evolution of player performance in the NHL. This scarcity made it difficult to ground our study in existing literature and may have impacted the depth and context of our analysis. The absence of a broader scholarly conversation poses a risk to the study's validity.

3. **Analytical Skills:**

   Challenges were encountered in the application of analytical skills. Despite efforts to employ various statistical methods, the complexity of hockey player performance

introduces the possibility of oversights or missed opportunities for more nuanced analyses. This limitation raises concerns about the study's overall validity, emphasizing the need for ongoing skill development and collaboration with experts in future research endeavors

10. Work Cited List

Diaz-Papkovich, Alexandr. “Data Mining the Play-By-Play: Assessing and Applying NHL Performance Metrics Using Statistical Methods. A thesis submitted to the Faculty of Graduate and Postdoctoral Affairs in partial fulfillment of the requirements for the degree of Master of Science in Probability and Statistics.” *Carlton University Ottawa, Canada, 2015.*

Brander, Egan, and Yeung *. " Estimating the Effects of Age on NHL Player Performance" *Sauder School of Business, University of British Columbia, Canada, August 2013.*

Global and US Sports Practice Leader (2023, January 24). 2023 Sports Industry Outlook. Deloitte United States. *https://www2.deloitte.com/us/en/pages/technology-media-and telecommunications/articles/sports-business-trends-disruption.html*

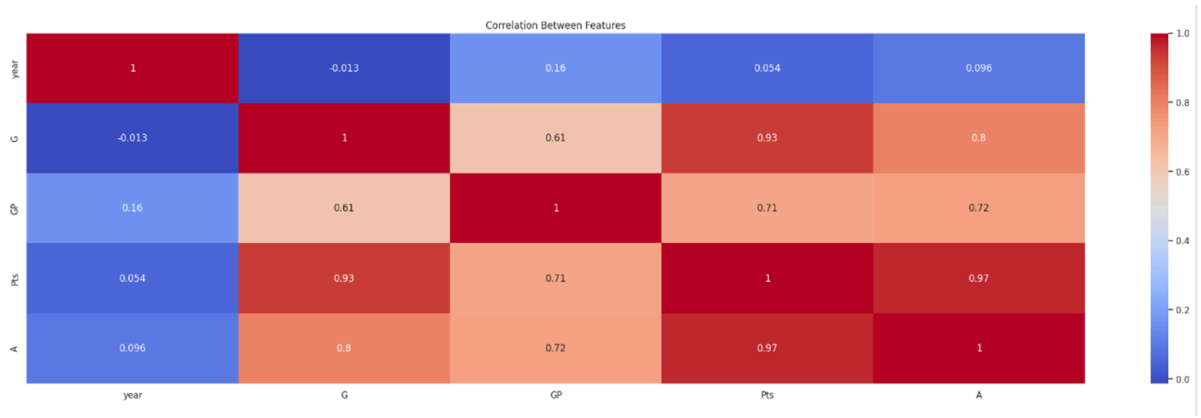
NHL EDGE advanced stats section brings fans closer to game *https://www.nhl.com/news/nhl-edge-advanced-stats-section-brings-fans-closer-to-game*

NHL Analytics Measure Player and Team Value (2022, June 2024) The Athletic. *https://theathletic.com/3377555/2022/06/22/nhl-analytics-player-team-value/*

Data Mining the Play-by-Play: Assessing and Applying NHL Performance Metrics Using Statistical Methods. Carleton University Library repository https://repository.library.carleton.ca/concern/etds/sf268576g

11. Appendix

Appendix 1.



Appendix 2.



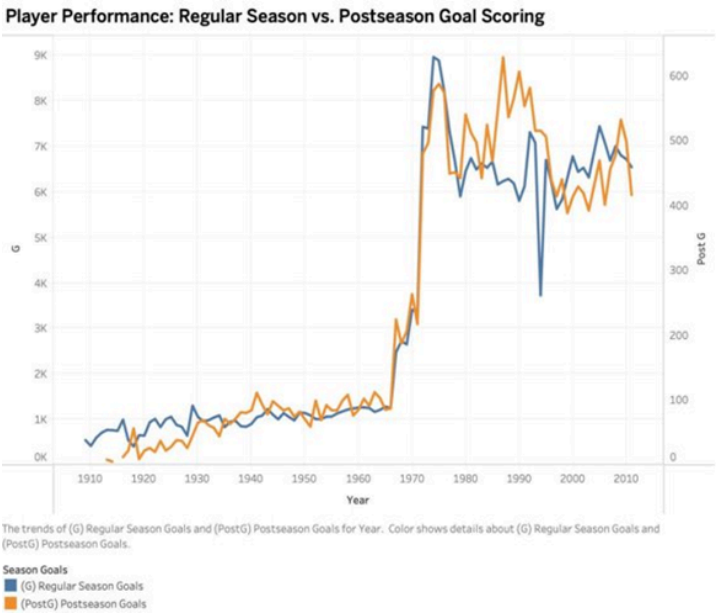
Appendix 3:

Descriptive statistics of major perferformance Indicator and Age					
Game Played		**Goals**		**Age**	
Mean	249.795	Mean	44.038	Mean	25.522
Standard Error	3.620	Standard Error	1.009	Standard Error	0.374
Median	101.500	Median	6.000	Median	26.000
Mode	1.000	Mode	0.000	Mode	0.000
Standard Deviation	313.960	Standard Deviation	87.501	Standard Deviation	32.419
Sample Variance	98571.015	Sample Variance	7656.511	Sample Variance	1050.995
Kurtosis	2.026	Kurtosis	16.668	Kurtosis	3191.426
Skewness	1.549	Skewness	3.508	Skewness	54.205
Range	2186.000	Range	975.000	Range	1944.000
Minimum	0.000	Minimum	0.000	Minimum	0.000
Maximum	2186.000	Maximum	975.000	Maximum	1944.000
Sum	1878459.000	Sum	331167.000	Sum	191922.000
Count	7520.000	Count	7520.000	Count	7520.000
Confidence Level(95%)	7.097	Confidence Level(95%)	1.978	Confidence Level(95%)	0.733

Appendix 4.

Height Vs Weight Analysis						
SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.9226					
R Square	0.8511					
Adjusted R Square	0.8511					
Standard Error	4.6570					
Observations	7520.0000					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	1.0000	932025.3842	932025.3842	42974.3716	0.0000	
Residual	7518.0000	163049.8965	21.6879			
Total	7519.0000	1095075.2807				
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	13.0032	0.2797	46.4901	0.0000	12.4549	13.5515
Weight	0.3088	0.0015	207.3026	0.0000	0.3058	0.3117

Appendix 5.



Appendix 6.

